

# Data Catalog Implementation Plan

Organization: Company A - Saudi Healthcare Authority  
Document ID: DC-PLAN-002  
Version: 1.5 (Post-Implementation Review)  
Classification: INTERNAL  
Date: March 2024  
Total Pages: 18  
Status: COMPLETED

## DOCUMENT CONTROL

| Version | Date           | Author           | Changes                    | Approver             |
|---------|----------------|------------------|----------------------------|----------------------|
| 1.0     | June 2023      | Sara Al-Mohsen   | Initial plan               | Dr. Khalid Al-Otaibi |
| 1.2     | September 2023 | Sara Al-Mohsen   | Phase 2 updates            | Hassan Al-Ghamdi     |
| 1.4     | December 2023  | Ahmed Al-Qahtani | Phase 4 lessons            | Dr. Khalid Al-Otaibi |
| 1.5     | March 2024     | Ahmed Al-Qahtani | Post-implementation review | Dr. Khalid Al-Otaibi |

Document Owner: Data Management Office - Tools Team  
Review Cycle: Project document (Final version)  
Distribution: Project Team, Steering Committee, IT Operations

## APPROVAL SIGNATURES

| Role                   | Name                 | Signature | Date           |
|------------------------|----------------------|-----------|----------------|
| Project Sponsor        | Dr. Khalid Al-Otaibi | [Signed]  | March 15, 2024 |
| IT Operations Director | Mohammed Al-Zahrani  | [Signed]  | March 14, 2024 |
| Project Manager        | Sara Al-Mohsen       | [Signed]  | March 13, 2024 |

## TABLE OF CONTENTS

1. PROJECT CHARTER ..... 4

|                                      |    |
|--------------------------------------|----|
| 2. ARCHITECTURE DESIGN .....         | 6  |
| 3. IMPLEMENTATION PHASES .....       | 8  |
| 4. RESOURCE & BUDGET PLAN .....      | 12 |
| 5. RISKS & MITIGATION .....          | 14 |
| 6. LESSONS LEARNED .....             | 16 |
| 7. POST-IMPLEMENTATION METRICS ..... | 18 |
| APPENDIX A: SYSTEM INVENTORY .....   | 20 |

# 1. PROJECT CHARTER

## 1.1 Project Overview

| Attribute    | Details                                |
|--------------|----------------------------------------|
| Project Name | Enterprise Data Catalog Implementation |
| Project ID   | PROJ-2023-DC-001                       |
| Start Date   | June 1, 2023                           |
| End Date     | February 28, 2024                      |
| Duration     | 9 months                               |
| Status       | Completed Successfully                 |

## 1.2 Business Objectives

| Objective        | Description                        | Priority |
|------------------|------------------------------------|----------|
| Data Discovery   | Enable self-service data discovery | High     |
| Lineage Tracking | End-to-end data lineage visibility | High     |
| Compliance       | Support NDMO control requirements  | High     |
| Governance       | Centralize metadata management     | Medium   |
| Productivity     | Reduce time to find data by 70%    | Medium   |

## 1.3 Success Criteria

| Criterion | Target | Achieved |
|-----------|--------|----------|
|-----------|--------|----------|

|                        |                    |               |
|------------------------|--------------------|---------------|
| Systems Connected      | 58 systems         | 62 systems ✓  |
| Business Terms Defined | 5,000 terms        | 5,847 terms ✓ |
| Active Users (Monthly) | 300 users          | 412 users ✓   |
| Lineage Coverage       | Regulatory reports | 100% ✓        |
| User Satisfaction      | 80%                | 87% ✓         |

## 1.4 Scope Definition

### In Scope

| Item           | Description                                         |
|----------------|-----------------------------------------------------|
| Platform       | Informatica Enterprise Data Catalog (EDC)           |
| Data Sources   | All production databases and data warehouses        |
| Metadata Types | Technical, business, operational                    |
| Lineage        | ETL-based and API-based lineage                     |
| Users          | All data consumers (analysts, stewards, developers) |

### Out of Scope

| Item                | Rationale                             |
|---------------------|---------------------------------------|
| Data Quality        | Separate project (IDQ Implementation) |
| Master Data         | Separate initiative planned for 2025  |
| Development Systems | Not required for production catalog   |

## 1.5 Key Stakeholders

| Stakeholder          | Role                | Involvement               |
|----------------------|---------------------|---------------------------|
| Dr. Khalid Al-Otaibi | Project Sponsor     | Strategic direction       |
| Sara Al-Mohsen       | Project Manager     | Day-to-day management     |
| John Smith           | Technical Architect | Architecture design       |
| Fatima Al-Zahrani    | Business Lead       | Requirements and adoption |
| IT Operations        | Infrastructure      | Platform support          |

## 2. ARCHITECTURE DESIGN

### 2.1 Infrastructure Specifications

| Component           | Specification                   | Quantity  |
|---------------------|---------------------------------|-----------|
| Application Servers | Dell PowerEdge R750             | 3 nodes   |
| CPU                 | Intel Xeon Gold 6348 (28 cores) | Per node  |
| RAM                 | 256 GB DDR4                     | Per node  |
| Storage             | Dell EMC Unity 480F (5 TB SSD)  | 1 array   |
| Operating System    | Red Hat Enterprise Linux 8.6    | All nodes |
| Network             | 10 Gbps dedicated VLAN          | Cluster   |

### 2.2 Software Components

| Component            | Version | Purpose               |
|----------------------|---------|-----------------------|
| Informatica EDC      | 10.5.2  | Core catalog platform |
| Informatica LDMADMIN | 10.5.2  | Model repository      |
| PostgreSQL           | 14.2    | Metadata database     |
| Apache Solr          | 8.11    | Search indexing       |
| Apache Kafka         | 3.2     | Event streaming       |

### 2.3 Network Architecture

| Zone        | Components          | Access             |
|-------------|---------------------|--------------------|
| DMZ         | Load balancer, WAF  | External users     |
| Application | EDC nodes, Solr     | Internal           |
| Database    | PostgreSQL cluster  | Application only   |
| Integration | Scanner jump server | Restricted sources |

### 2.4 Integration Pattern

| Integration    | Method           | Description                     |
|----------------|------------------|---------------------------------|
| Source Systems | JDBC/ODBC        | Direct database scanning        |
| ETL Tools      | Native connector | Informatica PowerCenter lineage |
| BI Tools       | REST API         | Tableau, Power BI metadata      |
| Authentication | LDAP/SAML        | Active Directory SSO            |
| Monitoring     | SNMP/Syslog      | Splunk integration              |

## 2.5 High Availability Design

| Component   | HA Method               | RTO    |
|-------------|-------------------------|--------|
| Application | Active-Active cluster   | 5 min  |
| Database    | Synchronous replication | 0 min  |
| Search      | Distributed Solr        | 2 min  |
| Storage     | RAID 10 + snapshots     | 15 min |

## 3. IMPLEMENTATION PHASES

### Phase 1: Foundation (Weeks 1-4)

#### *Week 1: Server Provisioning*

| Task                  | Owner        | Status    |
|-----------------------|--------------|-----------|
| Server procurement    | IT Ops       | Completed |
| Rack installation     | DC Team      | Completed |
| Network configuration | Network Team | Completed |
| OS installation       | IT Ops       | Completed |

#### *Week 2: Software Installation*

| Task                    | Owner          | Status    |
|-------------------------|----------------|-----------|
| EDC 10.5.2 installation | Informatica PS | Completed |
| PostgreSQL setup        | DBA Team       | Completed |

|                    |        |           |
|--------------------|--------|-----------|
| Solr configuration | IT Ops | Completed |
| Kafka deployment   | IT Ops | Completed |

Week 3: Security Hardening

| Task                         | Owner         | Status    |
|------------------------------|---------------|-----------|
| SSL certificate installation | Security Team | Completed |
| Firewall rules               | Network Team  | Completed |
| AD integration               | IAM Team      | Completed |
| Vulnerability scan           | Security Team | Completed |

Week 4: Connection Testing

| Source System      | Connection Type | Test Result |
|--------------------|-----------------|-------------|
| Oracle EHR         | JDBC Thin       | Pass        |
| SQL Server HR      | JDBC            | Pass        |
| PostgreSQL DW      | JDBC            | Pass        |
| Teradata Analytics | ODBC            | Pass        |

Phase 2: Ingestion (Weeks 5-12)

Scanning Strategy

| Group        | Systems               | Scan Frequency | Priority |
|--------------|-----------------------|----------------|----------|
| A - Critical | EHR, ERP, Payroll     | Daily          | High     |
| B - High     | CRM, SCM, LIS         | Weekly         | Medium   |
| C - Medium   | Departmental DBs      | Bi-weekly      | Medium   |
| D - Low      | File Shares, Archives | Monthly        | Low      |

Systems Scanned by Week

| Week   | Systems Added          | Cumulative Total |
|--------|------------------------|------------------|
| Week 5 | Oracle EHR, SAP ERP    | 2                |
| Week 6 | Oracle Payroll, SQL HR | 4                |

|            |                            |    |
|------------|----------------------------|----|
| Week 7     | Philips PACS, Sunquest LIS | 6  |
| Week 8     | Teradata DW, Oracle BI     | 8  |
| Week 9-10  | 12 departmental systems    | 20 |
| Week 11-12 | 42 additional systems      | 62 |

Challenges Encountered

| Challenge            | Impact           | Resolution             |
|----------------------|------------------|------------------------|
| EHR profile failure  | Delayed scan     | Implemented sampling   |
| Network timeout      | Slow scans       | Dedicated scan VLAN    |
| Character encoding   | Garbled metadata | UTF-8 normalization    |
| Large tables (450GB) | Memory issues    | Row sampling (1M rows) |

Phase 3: Enrichment (Weeks 13-20)

AI Tagging (CLAIRE Engine)

| Tag Type       | Auto-Tagged | Human Verified | Precision |
|----------------|-------------|----------------|-----------|
| PII Fields     | 2,847       | 500 sample     | 92%       |
| PHI Fields     | 1,923       | 300 sample     | 89%       |
| Financial      | 856         | 200 sample     | 94%       |
| Reference Data | 1,245       | 100 sample     | 96%       |

Business Glossary Import

| Domain       | Terms Imported | Owner               |
|--------------|----------------|---------------------|
| Clinical     | 2,134          | Dr. Ahmed Al-Rashid |
| Financial    | 1,456          | Sara Al-Mohsen      |
| HR           | 823            | Mohammed Al-Qahtani |
| Supply Chain | 567            | Khalid Al-Harbi     |
| IT           | 412            | Faisal Al-Salem     |
| Research     | 455            | Dr. Nadia Al-Otaibi |

Data Steward Assignments

| Week       | Stewards Onboarded | Domains Assigned  |
|------------|--------------------|-------------------|
| Week 13-14 | 15                 | Clinical, Finance |
| Week 15-16 | 12                 | HR, Supply Chain  |
| Week 17-18 | 10                 | IT, Research      |
| Week 19-20 | 8                  | Cross-domain      |

## Phase 4: Lineage Building (Weeks 21-26)

### Lineage Sources

| Source Type       | Tool                 | Coverage |
|-------------------|----------------------|----------|
| ETL Mappings      | PowerCenter          | 100%     |
| Stored Procedures | PL/SQL Parser        | 85%      |
| Views             | Database metadata    | 100%     |
| Java Applications | Custom Resource      | 60%      |
| APIs              | Manual documentation | 40%      |

### Priority Lineage Paths

| Path              | Start      | End             | Status   |
|-------------------|------------|-----------------|----------|
| Patient Journey   | Epic EHR   | MOH Report      | Complete |
| Financial Close   | Oracle EBS | Board Dashboard | Complete |
| HR Reporting      | SAP HR     | Ministry Report | Complete |
| Clinical Research | LIS        | Research DW     | Complete |

### Lineage Gaps

| Gap          | Reason            | Mitigation                 |
|--------------|-------------------|----------------------------|
| Java code    | No native scanner | Custom resource definition |
| Excel macros | Complex logic     | Manual documentation       |
| Legacy COBOL | End of life       | Deferred                   |

## Phase 5: Rollout & Training (Weeks 27-30)

Training Delivery

| Course           | Format    | Attendees | Completion |
|------------------|-----------|-----------|------------|
| Catalog Basics   | eLearning | 412       | 95%        |
| Advanced Search  | Workshop  | 156       | 89%        |
| Steward Training | Classroom | 45        | 100%       |
| Admin Training   | Hands-on  | 8         | 100%       |

Adoption Activities

| Activity          | Description             | Outcome          |
|-------------------|-------------------------|------------------|
| Treasure Hunt     | Gamified data discovery | 234 participants |
| Office Hours      | Weekly Q&A sessions     | High engagement  |
| Champions Network | Super users             | 12 champions     |
| Newsletter        | Monthly tips            | 85% open rate    |

4. RESOURCE & BUDGET PLAN

4.1 Implementation Team

| Role                    | Name                    | Allocation | Duration |
|-------------------------|-------------------------|------------|----------|
| Project Manager         | Sara Al-Mohsen          | 50%        | 9 months |
| Technical Architect     | John Smith (Contractor) | 100%       | 6 months |
| Infrastructure Engineer | Khalid Al-Salem         | 20%        | 4 months |
| Security Analyst        | Ahmed Al-Rashid         | 10%        | 2 months |
| Business Analysts       | 2 FTEs                  | 50%        | 6 months |
| Data Stewards           | 15 Business SMEs        | 5 hrs/week | 4 months |

4.2 Budget Summary

| Category | Budgeted (SAR) | Actual (SAR) | Variance |
|----------|----------------|--------------|----------|
|----------|----------------|--------------|----------|

|                       |         |         |         |
|-----------------------|---------|---------|---------|
| Software Licenses     | 300,000 | 280,000 | -20,000 |
| Hardware              | 100,000 | 120,000 | +20,000 |
| Professional Services | 400,000 | 400,000 | 0       |
| Training              | 50,000  | 30,000  | -20,000 |
| Contingency           | 50,000  | 0       | -50,000 |
| Total                 | 900,000 | 830,000 | -70,000 |

### 4.3 Budget Variance Analysis

| Variance    | Amount  | Explanation                    |
|-------------|---------|--------------------------------|
| Licenses    | -20,000 | Negotiated volume discount     |
| Hardware    | +20,000 | Additional RAM for performance |
| Training    | -20,000 | Developed internal eLearning   |
| Contingency | -50,000 | No major issues materialized   |

## 5. RISKS & MITIGATION

### 5.1 Risk Register

| Risk ID | Risk Description                 | Likelihood | Impact | Rating |
|---------|----------------------------------|------------|--------|--------|
| R001    | Source system performance impact | Medium     | High   | High   |
| R002    | Low user adoption                | Medium     | Medium | Medium |
| R003    | Data quality issues in metadata  | High       | Medium | High   |
| R004    | Integration complexity           | Medium     | Medium | Medium |
| R005    | Resource availability            | Low        | High   | Medium |

### 5.2 Mitigation Actions

| Risk ID | Mitigation                           | Owner  | Status      |
|---------|--------------------------------------|--------|-------------|
| R001    | Schedule scans 2-5 AM                | IT Ops | Implemented |
| R002    | Catalog required for access requests | PM     | Implemented |

|      |                             |            |             |
|------|-----------------------------|------------|-------------|
| R003 | Steward review process      | Governance | Implemented |
| R004 | Phased integration approach | Architect  | Implemented |
| R005 | Cross-training team members | PM         | Implemented |

## 5.3 Issues Log

| Issue                 | Date    | Resolution    | Status |
|-----------------------|---------|---------------|--------|
| EHR scan timeout      | Week 6  | Row sampling  | Closed |
| AD group sync delay   | Week 8  | Scheduled job | Closed |
| Solr index corruption | Week 14 | Rebuild index | Closed |
| Lineage display bug   | Week 22 | Vendor patch  | Closed |

## 6. LESSONS LEARNED

### 6.1 What Went Well

| Area            | Lesson                          | Recommendation                |
|-----------------|---------------------------------|-------------------------------|
| Sponsorship     | CDO involvement critical        | Maintain executive engagement |
| Training        | Gamification increased adoption | Use for future rollouts       |
| Phased approach | Reduced risk                    | Continue incremental delivery |
| Steward network | Business ownership key          | Invest in steward program     |

### 6.2 What Could Improve

| Area              | Lesson                    | Recommendation                  |
|-------------------|---------------------------|---------------------------------|
| Java lineage      | Underestimated complexity | Plan custom development earlier |
| User expectations | Gap in data quality       | Set clear scope boundaries      |
| Documentation     | Inconsistent quality      | Standardize templates           |
| Testing           | Insufficient UAT time     | Extend UAT phase                |

### 6.3 Best Practices Identified

| Practice           | Description                   |
|--------------------|-------------------------------|
| Steward champions  | Identify super users early    |
| Scan scheduling    | Off-peak hours mandatory      |
| Glossary ownership | Business must own definitions |
| Lineage validation | Steward sign-off required     |

## 7. POST-IMPLEMENTATION METRICS

### 7.1 Adoption Metrics (March 2024)

| Metric               | Target | Actual | Trend |
|----------------------|--------|--------|-------|
| Monthly Active Users | 300    | 412    | ↑     |
| Daily Searches       | 500    | 687    | ↑     |
| Assets Viewed        | 10,000 | 14,234 | ↑     |
| Access Requests      | 100    | 156    | ↑     |
| Questions Asked      | 50     | 78     | ↑     |

### 7.2 Quality Metrics

| Metric                 | Target | Actual   |
|------------------------|--------|----------|
| Metadata Completeness  | 80%    | 87%      |
| Business Term Coverage | 70%    | 82%      |
| Lineage Accuracy       | 90%    | 94%      |
| Steward Response Time  | 2 days | 1.5 days |

### 7.3 Business Value

| Benefit            | Measurement         | Result        |
|--------------------|---------------------|---------------|
| Time to find data  | Before/After survey | 70% reduction |
| Report development | Project tracking    | 40% faster    |

|                        |                   |                          |
|------------------------|-------------------|--------------------------|
| Compliance preparation | Audit feedback    | 2 days → 4 hours         |
| Duplicate analysis     | Impact assessment | 15 duplicates identified |

# APPENDIX A: SYSTEM INVENTORY

## A.1 Connected Systems Summary

| Category   | System Count | Key Systems                          |
|------------|--------------|--------------------------------------|
| Clinical   | 12           | Epic EHR, Philips PACS, Sunquest LIS |
| Financial  | 8            | Oracle EBS, SAP FICO                 |
| HR         | 5            | SAP HR, Payroll                      |
| Analytics  | 6            | Teradata, Tableau, Power BI          |
| Operations | 15           | SCM, Facilities, Fleet               |
| Research   | 8            | REDCap, Research DW                  |
| Other      | 8            | File shares, legacy                  |

Document End

Project Steering Committee Sign-off  
Company A - Saudi Healthcare Authority  
Email: datacatalog@companya.gov.sa  
Version 1.5 | March 2024